Curriculum Vitae

Nabanita Karmakar, M.Sc.
Gender: Female
Date of birth: 30 September 1998
Home address: Shibpur, Howrah,
West Bengal, India

ORCID: 0009-0002-0736-9540
Scopus ID: 59195076700
Phone: (+91) 8337042515
Email: k.nabanita@iitg.ac.in,
nabanitakarmakar2026@gmail.com

Educational Qualification

Ph.D.

2022 (January) – Thesis submitted (September 16, 2026), Department of Mathematics, Indian Institute of Technology Guwahati, Guwahati, Assam, India.

- **CGPA: 8.36/10**
- *Thesis: Impact of Various Porous Breakwaters on Wave Scattering and Mitigation of Forces on a Rigid Tunnel Placed in a Sea Possessing Different Types of Bottom Topography*
- Thesis Advisor: Prof. Swaroop Nandan Bora

M.Sc.

2021, Applied Mathematics, Department of Mathematics, Indian Institute of Engineering Science and Technology, Shibpur, West Bengal, India.

- **CGPA: 9.92/10**
- *Project 1: Nonlinear Dynamics and its Application in Lorentz like Model* (Project Supervisor: Prof. Pritha Das)
- *Project 2: Stability Analysis of Predator-Prey Model with Stage-Structure and Anti-Predator Behavior* (Project Supervisor: Prof. Pritha Das)
- *Project 3: Eigenvalue Approach of Solution of Coupled Non-Homogeneous Differential Equations* (Project Supervisor: Prof. Smita Pal (Sarkar))
- *Project 4: Fractional Calculus and its Application in Thermoelasticity* (Project Supervisor: Prof. Smita Pal (Sarkar))

B.Sc.

2019, Mathematics (66.12%)
Department of Mathematics, Narasinha Dutt College, University of Calcutta, West Bengal, India.

Higher Secondary

2016, Science (85.6%)
Shibpur Hindu Girls High School, West Bengal Council of Higher Secondary Education, West Bengal, India.

Secondary

2014 (84.86%)
Shibpur Hindu Girls High School, West Bengal Board of Secondary Education, West Bengal, India.

Awards

- Ganesh Chandra Mitra Memorial Medal - for securing 1st position among all disciplines postgraduate students, 2021, Indian Institute of Engineering Science and Technology, Shibpur, West Bengal, India.
- Arun Chandra Mitra Memorial Medal - for securing 1st position among all Master of Science students, 2021, Indian Institute of Engineering Science and Technology, Shibpur, West Bengal, India.
- Prof. S. C. Dasgupta Gold Medal - for securing 1st position among all Master of Science in Applied Mathematics students, 2021, Indian Institute of Engineering Science and Technology, Shibpur, West Bengal, India.

Examinations Qualified

• Qualified JAM, 2019 • Qualified GATE, 2021 • Qualified CSIR NET (LS), June 2021

Areas of Interest

• Fluid dynamics (Water wave mechanics, Wave-structure interaction) • Applications of differential equations • Numerical analysis • Linear and nonlinear water waves • Computational fluid dynamics

Courses Undertaken during PhD

• Differential Equations • Real Analysis • Linear Algebra • Fourier Analysis • Numerical Methods • Fluid Dynamics • Machine Learning

Experience as a Teaching Assistant

Acted as Tutor to Bachelor courses in calculus, linear algebra, differential equations and complex analysis; to Masters courses in differential equations.

Technical Skills

• MATLAB • Python • LaTeX

Research Goals

My research goal is to develop mathematical and computational models that improve our understanding of water wave-structure interaction and contribute to the safe and sustainable design of offshore and coastal infrastructure. I am particularly interested in investigating wave scattering, fluid–structure interaction, and wave mitigation using porous structures under diverse marine environments. In the future, I aim to extend my work to more realistic and complex problems, including nonlinear wave effects, hydroelasticity of floating structures, and renewable offshore energy systems, while combining analytical and numerical techniques to address practical challenges in marine engineering. However, I am open to work in any other areas of research which align with my prevailing knowledge.

Publications

1. **Nabanita Karmakar** and Swaroop Nandan Bora (2026), Wave scattering by a composite porous breakwater mounted on a rigid pedestal and placed in front of a rigid tunnel over a porous sea-bed, *Ocean Engineering*, 358:125695 (20 pages).
2. **Nabanita Karmakar** and Swaroop Nandan Bora (2026), Effect of two thin porous barriers and a porous sea-bed on water wave scattering and force mitigation in protecting a rectangular rigid tunnel, *Journal of Ocean Engineering and Marine Energy*, 12:1165-1190.
3. **Nabanita Karmakar** and Swaroop Nandan Bora (2025), The impact of a composite rectangular porous breakwater in the scattering of water waves by a submerged rectangular tunnel, *International Journal of Applied Mechanics*, 17(8):2550056 (30 pages).
4. **Nabanita Karmakar** and Swaroop Nandan Bora (2024), Role of a thin porous vertical barrier and a step-type sea-bed in reflecting waves and mitigating wave forces acting on a submerged tunnel, *Ocean Engineering*, 310:118568 (16 pages).

Conference Presentation

1. Role of a thin vertical porous barrier and a step-type sea-bed in reflecting waves and mitigating wave forces acting on a submerged tunnel, *69th Annual Congress of Indian Society of Theoretical and Applied Mechanics (ISTAM)*, Christ University, Bengaluru, India, December 19-21, 2024.
2. Reflection of waves and reduction of wave forces acting on a submerged tunnel by a step-type sea-bed and a thin porous vertical barrier, *National Conference on Theoretical and Numerical Approaches in Water Wave Mechanics*, Vellore Institute of Technology, Vellore, Tamil Nadu, India, December 16-17, 2024.

Conference and Workshop Participation

1. *7th International Conference on Ship and Offshore Technology (ICSOT)*, Indian Institute of Technology Kharagpur, West Bengal, India, December 8–9, 2023.
2. *International Workshop on Mathematical and Computational Methods in Science and Engineering (IWMCMSE)* Department of Mathematics, Indian Institute of Technology Ropar, Punjab, India, June 2-13, 2023 (Online).

References

1. Prof. Swaroop Nandan Bora, Thesis Supervisor, Department of Mathematics, Indian Institute of Technology Guwahati, Email id: swaroop@iitg.ac.in, Office Phone No.: +91 361 258 2604.
2. Prof. Durga Charan Dalal, Chairman, Doctoral Committee, Department of Mathematics, Indian Institute of Technology Guwahati, Email id: durga@iitg.ac.in, Office Phone No.: +91 361 258 2615.
3. Prof. Natesan Srinivasan, Member, Doctoral Committee, Department of Mathematics, Indian Institute of Technology Guwahati, Email id: natesan@iitg.ac.in, Office Phone No.: +91 361 258 2613.

4. Prof. Rajen Kumar Sinha, Member, Doctoral Committee, Department of Mathematics, Indian Institute of Technology Guwahati, Email id: rajen@iitg.ac.in, Office Phone No.: +91 361 258 2612.